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Project :- Spread locator

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Statistical Distribution Concepts For Spread Locator

→ Introduction.

This document provides a brief explanation of the statistical used in the Spread Locator project. These concepts were applied throughout the Jupyter Notebook to analyse customer transaction data, identify suitable probability distributions, and generate business insights using Python.

1. Statistical Distribution

A statistical distribution how data values are spread across a dataset. It helps us understand patterns such as whether the data is normally distributed, skewed, or clustered around certain values. Statistical distributions are important because they help analysts choose the correct statistical methods for analysing data.

2. Q-Q Plot

A Q-Q (Quantile - Quantile) Plot is a graph used to compare a dataset with a theoretical Normal distribution. If the plotted points closely follow the reference line, the data is approximately Normally distributed. Large deviations from the line indicate that another distribution may better represent the data.

3. Discrete and Continuous Distributions

A Discrete Distribution represents countable values, such as the number of customer transactions.

A Continuous Distribution represents measurable values, such as transaction amounts.

Choosing the correct distribution depends on the type of data being analysed.

4. Bernoulli Distribution

The Bernoulli Distribution models an experiment with only two possible outcomes, such as Success or Failure. In this project, it was used to analyse transaction status.

5. Binomial Distribution

The Binomial Distribution calculates the probability of obtaining a certain number of successes in a fixed number of independent trials. It extends the Bernoulli Distribution by analysing multiple events instead of just one.

6. log-Normal Distribution

A log-Normal Distribution is used for positive, right-skewed data. Financial data and transactions amounts often follow this distribution because most values are small while only a few are very large.

7. Power law Distribution

A Power law distribution describes situations where a small number of observations account for a large proportion of the total outcome. In business, this often means that a few customers generate a large percentage of total sales.

8. Box-Cox Transformation

The Box-Cox Transformation is used to reduce skewness and make data more Normally distributed. It improves the performance of many statistical methods by creating a more balanced dataset.

9. Poisson Distribution.

The Poisson Distribution models the number of events occurring within a fixed period of time. It is commonly used for analysing counts, such as the number of customer transactions per day.

10. Z-score Probability

A Z-score measures how far a value is from the mean in terms of standard deviations. It is useful for identifying unusual observations (outliers) and calculating probabilities.

11. Probability Density Function (PDF) and Cumulative Distribution Function (CDF)

The Probability Density Function (PDF) shows how data values are distributed.

The Cumulative Distribution Function (CDF) shows the probability that a value is less than or equal to a specific value.

Both functions help analysts understand the probability behaviour of continuous data.